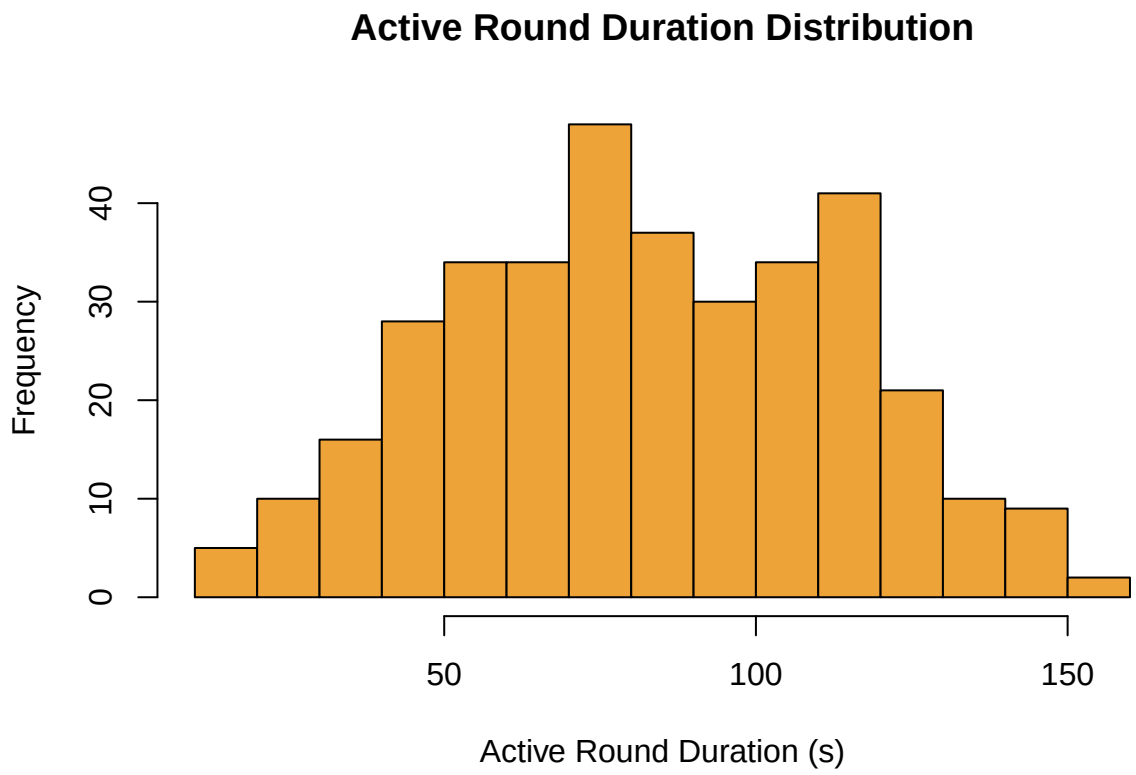
Descriptive Analysis Report

2026-10-28

Introduction:

The goal of this study is to analyze data from professional CS2 matches to find useful statistics for improving gameplay. I retrieved my data from the OpenCS2 dataset that contains statistics from professional CS2 matches. The sampling units are single rounds chosen randomly from a match. My final sample size is 359 rows. Of the 16,527 rows, 16168 rows were removed to use only 1 round from each match and to remove rows with missing data. One research question is to find how round duration associates with the amount of time until the first kill. The other research question is to find how the round duration associates with the winning side of the round. The round duration used for this analysis was recalculated to exclude the buy period at the beginning of the round (active round duration).

Descriptive Analysis of Response Variable

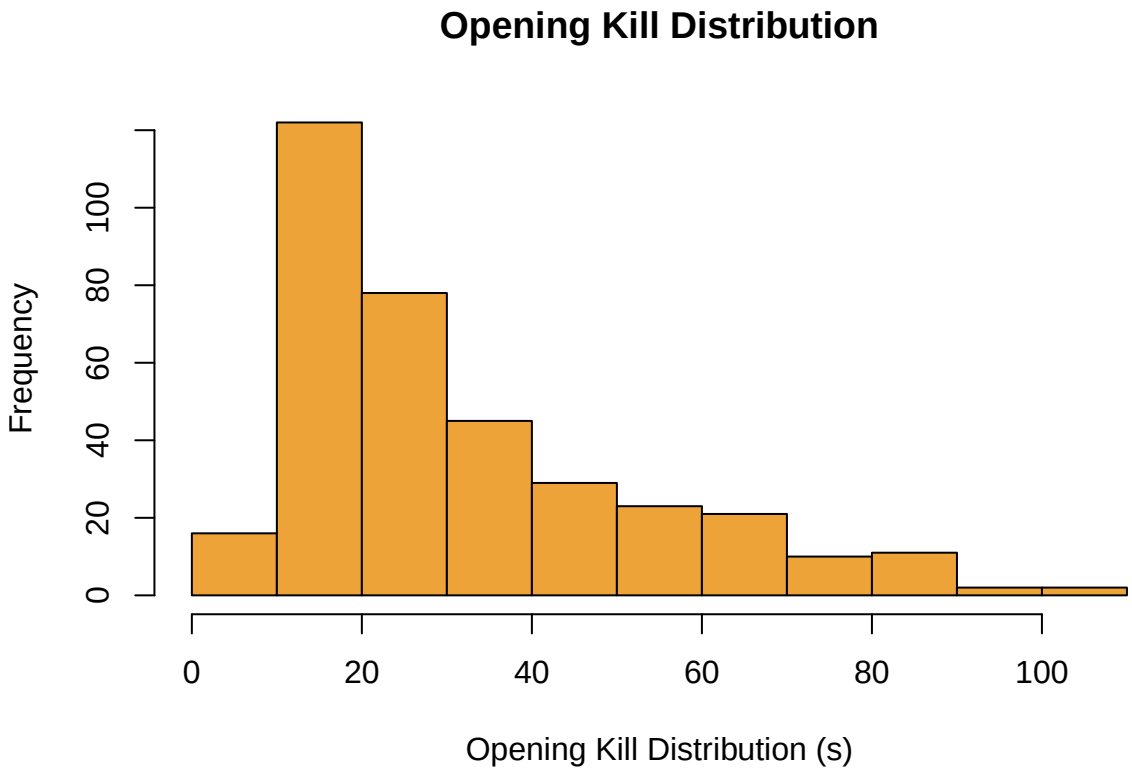



```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   18.41   59.35   80.92   82.68  107.87  153.98
```

```
## [1] 30.78224
```

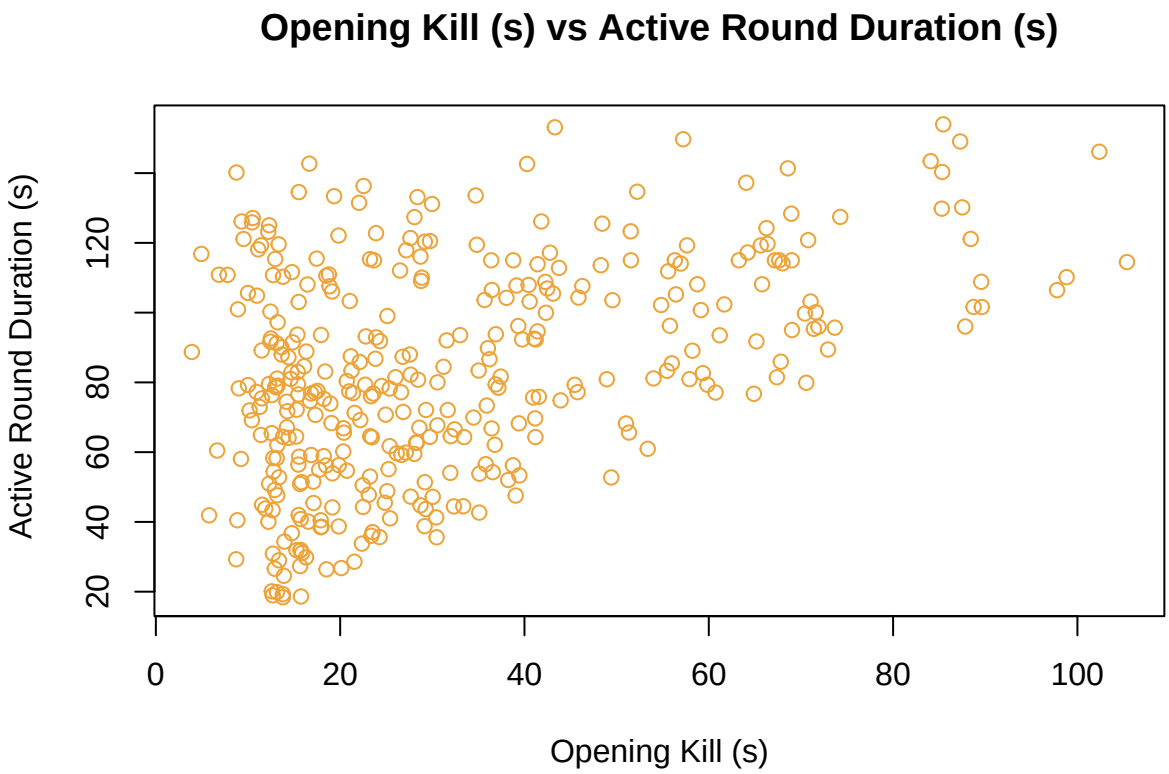
Active Round Duration appears to have a mostly normal distribution with a mean of 82.68 seconds and median of 80.92 seconds. It also has a standard deviation of 30.78 seconds.

Descriptive Analysis of Numeric Explanatory Variable



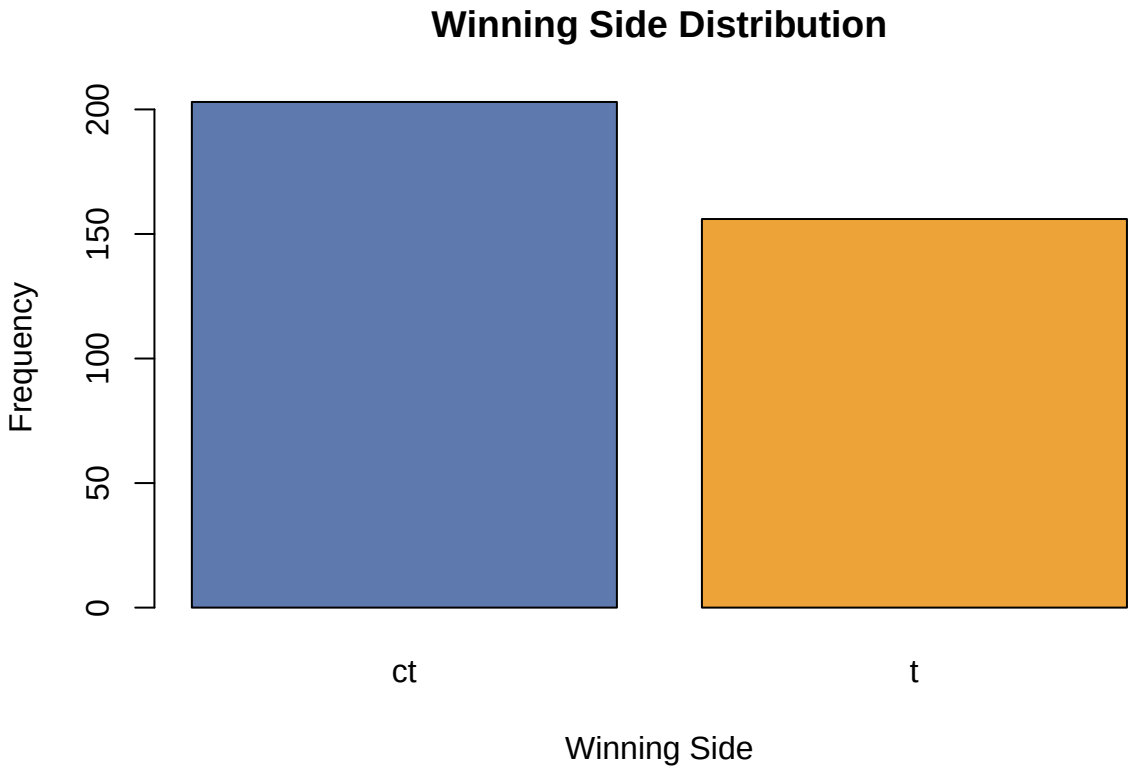

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   3.906  15.500  25.125  31.990  41.320 105.375
```

```
## [1] 21.22668
```



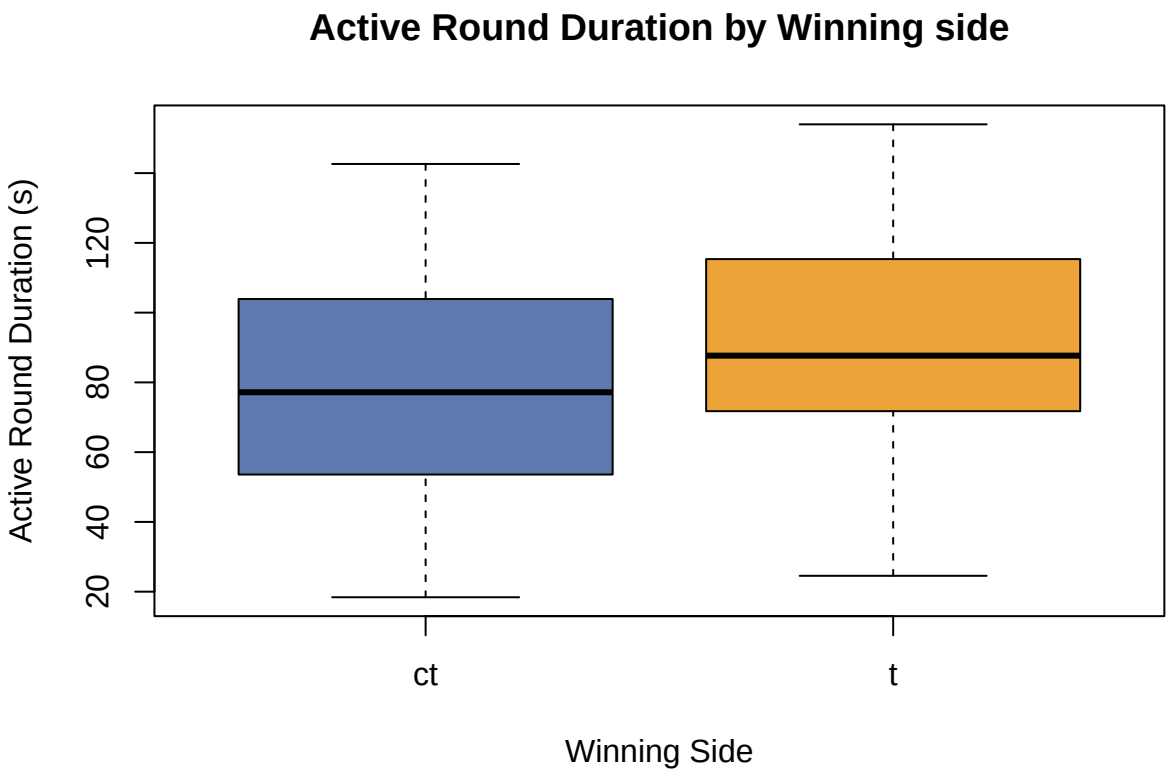
Opening kill has a right-skewed distribution with a mean of 31.99 seconds and a median of 25.13 seconds. It also has a standard deviation of 21.23 seconds. It has an IQR of 25.82 seconds and possibly some outliers. The scatterplot displays a positive association between the opening kill and the active round duration.

Investigation of Categorical Explanatory Variable




```
##
##  ct   t
## 203 156
```

```
##
##        ct         t
## 0.5654596 0.4345404
```



The barplot shows that CT had a greater win amount of 203 wins (56.55%) than T had with 156 wins (43.45%). The grouped boxplot displays that CT and T have a similar spread, with T having a slightly higher median.